

Arul Annamalai

Frontend Software Engineer

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LinkedIn | GitHub | Portfolio | LeetCode

Professional Summary

Frontend-focused Software Engineer with 2+ years of experience building scalable, high-performance web applications using React.js, Next.js, and TypeScript. Skilled in server-side rendering, component-based architecture, state management, and real-time systems (LiveKit, WebSockets), with a strong focus on code quality, performance optimization, and user experience.

Technical Skills

Frontend: ReactJS, NextJS, TypeScript, React Native, Redux, Zustand, HTML5, CSS3, Tailwind CSS

Languages: JavaScript (ES6+), TypeScript, Python

Backend: Node.js, Express.js, REST APIs, JWT Authentication, OAuth 2.0, WebSockets

Databases: MongoDB, MySQL, Redis

DevOps & Tools: Git, GitHub, Git-based CI/CD (Vercel), Docker (Basic), Postman, AWS (Basic), Firebase Cloud Messaging (FCM), LiveKit, Electron.js

Concepts: Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), Responsive Design, Web Performance Optimization

Experience

Frontend Software Engineer

Sep 2024 – Present

TeleCMI

Villupuram, India

- Architected a drag-and-drop **Call Flow Builder (React, TypeScript, Zustand)** for Piopiy, letting customers self-serve **IVR, AI voice agent**, and **PSTN** routing configuration without needing engineering support for changes.
- Selected and implemented **Zustand** for centralized state management in the Call Flow Builder, reducing prop drilling and simplifying state updates across deeply nested drag-and-drop components.
- Engineered communication features by integrating the **Connly SDK** and **Connly Video SDK**, delivering messaging, **LiveKit**-based audio/video calling, participant lifecycle management, media track synchronization, reconnection handling, and **CDN**-backed media uploads across **React Native** and **Electron.js**.
- Implemented secure authentication using Email OTP, TOTP, and **OAuth 2.0 (Google & GitHub Sign-In)**, improving user onboarding and strengthening access control across admin and customer-facing applications.
- Developed an offline-capable push notification system on **FCM** with **Redis**-backed state, ensuring reliable delivery and consistent sync across devices.
- Improved **Core Web Vitals** (LCP +35%, FCP +28%) and organic traffic (+20%) through lazy loading, code splitting, and asset optimization.
- Worked in **Agile/Scrum** (1–2 week sprints, daily standups) with code reviews before every release; mentored a junior teammate into joint ownership of the Piopiy/Connly codebase.

Projects

Quick Chat | React.js, Node.js, Express.js, MongoDB, Socket.IO

Live Demo | GitHub

- Developed a chat platform using the **MERN** stack, **Socket.IO**, and **Redux** for centralized state management; set up **Git-based CI/CD on Vercel** so every push to `main` auto-builds and deploys to production.
- Implemented **JWT** authentication, live user presence tracking, WebSocket-based messaging, and responsive UI integrated with REST APIs, achieving sub-**150ms** message latency for **500+** concurrent users.

NextBlog | Next.js, MongoDB, JWT, Tailwind CSS

Live Demo | GitHub

- Architected a full-stack blogging platform using **Next.js App Router** and **React Server Components**, implementing **server-side rendering** for SEO-optimized pages, with a **JWT**-authenticated admin CMS supporting full CRUD on articles via middleware-protected routes; deployed on **Vercel** with **Git-based CI/CD** for continuous delivery on every commit.
- Implemented live search, client/server pagination, and dynamic SEO metadata (**OpenGraph** tags, title templates), plus a **Tailwind CSS** UI with skeleton loading states and a REST-backed contact/inquiry module.

Achievements

- Solved **200+** Data Structures and Algorithms problems on **LeetCode**.
- Delivered production-ready features for an **AI-powered cloud communication platform**, including real-time messaging, audio/video calling, authentication, and workflow automation.

Education

Bachelor of Computer Applications (BCA)

2017 – 2020

Aringar Anna Government College, Villupuram

CGPA: 7.6

Certifications

Frontend Development – May 2025

JavaScript – Apr 2025

Problem Solving – Apr 2025

SQL – Feb 2024